

S11 – Sensitivity Analysis – Edge Deployment Thresholds

Purpose

The purpose of this analysis is to evaluate how variations in provisional clinical edge-deployment thresholds influence the classification of models as “edge-ready.” Specifically, the analysis investigates the sensitivity of the taxonomy’s Edge Readiness dimension to different latency and memory constraints, using the computational resource estimates reported in Tables 4 and 5 of the main manuscript.

Methodology

A range of deployment thresholds was systematically evaluated:

- **Latency thresholds:** 100 ms to 300 ms
- **Memory thresholds:** 50 MB to 150 MB

For each threshold combination, models were classified as edge-ready only if both latency and memory constraints were simultaneously satisfied.

The baseline governance thresholds were defined as:

- Latency $\leq$ 200 ms
- Memory $\leq$ 100 MB

Resource estimates for:

- GNN,
- Transformer, and
- Edge-Fusion variants

were obtained from the evaluation workstation reported in Table 4, whereas HIMS-CDI_cal measurements correspond to the Raspberry Pi 4 deployment described in Section 5.3.

Results

Latency Threshold (ms)	Memory Threshold (MB)	Models Passing	Change from Baseline
200 (+0%)	100 (+0%)	HIMS-CDI_cal only	–
100 (–50%)	100	None	–
300 (+50%)	100	HIMS-CDI_cal, Edge-Fusion (optimised)	+1
200	50 (–50%)	None	–
200	150 (+50%)	HIMS-CDI_cal, Edge-Fusion (optimised)	+1
300	150	HIMS-CDI_cal, Edge-Fusion (optimised)	+1

Additional Observations

- Edge-Fusion (optimised):

- Latency: 215 ms
- Memory: 98 MB

Accordingly, Edge-Fusion (optimised) fails the baseline latency threshold because 215 ms exceeds the 200 ms governance limit. However, it becomes edge-feasible when the latency constraint is relaxed to 300 ms.

Conversely:

- GNN and Transformer consistently fail all evaluated threshold combinations due to:
 - latency ≥ 310 ms, and
 - memory utilisation ≥ 420 MB.

Interpretation

The analysis demonstrates that the edge-ready status of HIMS-CDI_cal remains stable across all realistic threshold perturbations. In contrast, Edge-Fusion (optimised) exhibits borderline deployment feasibility and only becomes acceptable under relaxed operational conditions.

The persistent infeasibility of GNN and Transformer architectures highlights the practical limitations of computationally intensive deep learning models in constrained clinical edge environments.

This sensitivity pattern, therefore, aligns closely with the taxonomy's broader governance recommendation that HIMS-CDI MLR represents the safest default deployment option for edge-constrained healthcare settings.

Conclusion

The edge-readiness classification of HIMS-CDI_cal remains robust under $\pm 50\%$ threshold perturbation scenarios. Edge-Fusion (optimised) becomes operationally feasible only when governance constraints are relaxed, whereas GNN and Transformer architectures remain unsuitable for deployment on this class of clinical edge hardware. These findings reinforce the internal consistency of the taxonomy's Edge Readiness dimension, although future deployment trials using real IoT infrastructure remain necessary for external validation.